

Assignment 4 Report

Assignment 4 — Written Report

Project: Contactless Fingerprint Quality Assessment & Scoring Pipeline

Question 1: What threshold did you set for blur? How did you decide?

The Laplacian variance threshold was set to **10.0**, with a secondary normalization cap of 50.0 used when converting the raw score into the composite score's [0, 1] scale (a variance of 50 or higher is treated as "fully sharp").

Calibration methodology: I ran the blur metric against the synthetic and reference test captures spanning a range of motion levels. Sharp, steady captures consistently produced Laplacian variances between **25.0 and 150+**, while deliberately motion-blurred captures (simulated via a linear motion kernel, and cross-checked against handheld video frame extractions) consistently fell **below 8.0**. This left a clear separation band between roughly 8 and 25.

Justification: Setting the threshold at 10.0 sits close to the lower edge of that separation band. This is a deliberate choice: it is more tolerant of *soft focus* (slightly imperfect but still usable captures) while still reliably discarding images with *severe* motion blur that would meaningfully degrade downstream minutiae extraction. A stricter threshold (e.g. 20.0) would reject a non-trivial fraction of otherwise usable "soft" captures and increase user friction/re-capture rate without a proportional gain in matching accuracy.

Question 2: Which metric was hardest to implement correctly? What went wrong first?

Ridge Clarity and **ROI Completeness** were, by a clear margin, the two hardest metrics to get right.

ROI Completeness — what went wrong first: A naive fixed-intensity threshold (e.g. “foreground = pixels darker than 128”) fails as soon as the background tone happens to be similar to skin tone, or when uneven ambient lighting casts a shadow that the threshold misclassifies as part of the finger. The first implementation attempt produced wildly inconsistent `roi_fraction` values — sometimes reporting 90%+ “finger area” on images where the background dominated the frame, simply because the fixed threshold picked the wrong intensity class as foreground.

Fix: Switching to **Otsu’s method** removed the need to hand-pick an intensity constant — Otsu automatically finds the threshold that best separates the image’s bimodal histogram. The remaining ambiguity (which of the two resulting classes is “finger” vs. “background”) was resolved with a simple minority-class heuristic, which is a pragmatic approximation but not a substitute for true semantic segmentation in a production system (see the “Future Improvements” section of `README.md`).

Ridge Clarity — what went wrong first: An initial attempt used a plain Sobel edge-magnitude sum as a proxy for “ridge detail.” This produced **false positives**: high-contrast background clutter, skin folds, and even paper/desk textures behind the finger generated strong Sobel responses that had nothing to do with actual fingerprint ridges, inflating the clarity score for genuinely poor captures.

Fix: Replacing the generic edge detector with an **orientation-tuned Gabor filter bank** (averaged across four orientations) restricts the response to the specific periodic, directional texture that ridge-valley patterns actually exhibit, rather than any high-frequency content in general. Constraining this measurement to the segmented ROI (once available) further removes background-clutter false positives.

Question 3: What is NFIQ2? Why is a score designed for contact scanners not reliable for phone camera images?

NFIQ2 (NIST Fingerprint Image Quality 2) is the current industry standard, NIST-maintained open-source tool for quantifying fingerprint image quality on a normalized 0–100 scale, predictive of expected matcher performance. It was developed and calibrated against images collected on **contact-based optical and capacitive scanners**.

NFIQ2 is unreliable for contactless phone-camera captures for three main reasons:

1. **Acquisition physics gap.** NFIQ2's internal feature extractors encode assumptions native to frustrated total internal reflection (FTIR) optical scanners — a flat, pressure-applied contact surface, a fixed ~500 DPI resolution, and high-contrast black-and-white ridge rendering. None of these assumptions hold for a handheld phone camera capturing a 3D, unconstrained finger surface.
 2. **Perspective distortion and scale variation.** A phone capture's effective DPI varies continuously with capture distance, and the finger's curved 3D surface introduces non-linear ridge-spacing warping that a flat-scanner-trained quality model has never seen and cannot correctly interpret.
 3. **Texture and lighting differences.** Contactless images carry natural skin coloration, ambient shadows, and specular highlights — none of which appear in the pressure-flattened, illumination-controlled images NFIQ2 was calibrated on. These differences confuse NFIQ2's internal feature extraction, frequently producing artificially low quality scores even for objectively usable contactless captures — making NFIQ2 scores non-comparable, and unsuitable as a direct pass/fail gate, for this modality without substantial re-calibration or a purpose-built replacement (which is effectively what this pipeline is).
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Question 4: Name 3 other quality problems you'd add checks for in a real deployment.

1. **Perspective / pitch & yaw angle distortion.** A finger significantly tilted relative to the camera's optical axis produces non-uniform ridge spacing across the frame (near

edge foreshortened vs. far edge stretched), which degrades minutiae template consistency even when blur, brightness, and glare all pass. A dedicated angle-estimation check (e.g. via finger-silhouette aspect ratio or a lightweight pose estimator) should reject or request re-capture for extreme tilt.

2. **Inter-digital occlusion / multi-finger interference.** If an adjacent finger enters the capture bounding box, downstream segmentation and minutiae extraction modules can conflate ridge structures from two different fingers, corrupting the resulting template. A check that verifies the ROI mask corresponds to a *single, contiguous* connected component of plausible finger-shaped geometry would catch this before it reaches matching.
 3. **Distance / scale boundary check.** Too-far captures effectively downsample the finger below the resolution needed for reliable minutiae detection (well under the ~ 300 DPI-equivalent threshold used by contact scanners), while too-close captures fall outside the camera's minimum focal distance and produce optical blur that the Laplacian blur check may not fully distinguish from motion blur. An explicit scale/DPI estimation check (e.g. using a known reference object size, or a finger-width heuristic against the frame's field of view) would catch both failure modes directly rather than relying on blur/ROI checks to catch them indirectly.
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Question 5: If a rural agricultural worker's fingerprints are naturally worn and give consistently poor ridge clarity scores, what should the system do differently for them?

The underlying problem: Sustained manual labor and environmental exposure can physically wear down friction ridge relief, permanently lowering the achievable ridge-clarity signal for that individual — no amount of re-capture will produce a “sharp” ridge pattern that physically no longer exists at full contrast. A fixed, population-wide ridge-clarity threshold will systematically and unfairly reject this population, regardless of how well they perform the capture.

Recommended system adaptations:

1. **Dynamic, per-user adaptive thresholding.** During enrollment, detect consistently low ridge-clarity readings across multiple honestly well-composed attempts (i.e. all *other* metrics pass) and lower that specific user's ridge-clarity pass

threshold accordingly, ideally after applying localized contrast enhancement (e.g. **CLAHE** — Contrast Limited Adaptive Histogram Equalization) to recover as much genuine ridge signal as physically remains before deciding the threshold is truly unreachable.

2. **Multi-finger / alternative-finger fallback.** Automatically prompt the user to enroll or authenticate with a different digit (commonly the thumb or index finger) that may retain comparatively less wear than the finger most used for their specific manual tasks.
 3. **Multi-frame image fusion.** Capture a short burst (3–5 frames) during a single session and fuse them into one higher-SNR composite ridge map — averaging out sensor noise and transient specular artifacts can recover marginal ridge detail that no single frame captures cleanly.
 4. **Multimodal biometric fallback.** For users whose ridge structure is permanently and significantly degraded, the system should gracefully offer an alternative modality — e.g. facial recognition or iris scanning — rather than forcing repeated, inevitably failing fingerprint attempts, which is both a poor user experience and a form of biometric-system inequity that should be explicitly engineered against rather than left as an edge case.
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Summary

This pipeline demonstrates that a lightweight, sub-300ms, purely classical computer-vision QC gate (no deep learning required) can meaningfully triage contactless fingerprint captures before they reach expensive downstream models — while the accompanying analysis above outlines the concrete gaps (angle, occlusion, scale, and population-fairness) that a production-grade deployment would need to close beyond this assignment's scope.